

HAO PAN

Singapore

+65 8124 4894 | hao036@e.ntu.edu.sg | [linkedin.com/in/hao-pan-59193937a](https://www.linkedin.com/in/hao-pan-59193937a) | hao036.github.io

Summary

Data Agent / LLM Application Engineer building constrained, auditable analytics systems. Developed production Data Agent Skills, deterministic Text-to-SQL workflows, and field-level evaluations at Shopee; combines product diagnosis, failure-mode engineering, and human-in-the-loop design to turn business logic into reliable AI capabilities.

Education

Nanyang Technological University

Nov. 2025 – Jan. 2027

Master of Computing in Applied Artificial Intelligence, College of Computing and Data Science

Singapore

– Coursework: Agentic AI & Applications, Reinforcement Learning, and LLM Applications.

Suzhou University

Sep. 2019 – Jun. 2023

Bachelor of Engineering in Data Science and Big Data Technology

China

– **China National Scholarship** (Top 0.2% nationwide, 2022–2023).

Experience

Shopee Singapore

Mar. 2026 – Sep. 2026

AI Data Analyst Intern, Buyer & Marketing BI Team

Singapore

- Built and deployed **5 production Data Agent Skills** spanning PDP, OPC, Review, Cart, and a standardized Skill authoring template; migrated metric definitions, joins, filters, and calculation logic from the A/B testing platform into reusable, auditable runbooks.
- Designed deterministic execution boundaries: froze reference SQL, limited changes to **6 validated placeholders**, sharded one **group_id** per query, and required successful shard results to be concatenated without rewriting, optimization, or re-aggregation.
- Documented **15 explicit failure modes** across SQL rewriting, timeouts, partial results, live-table drift, and infrastructure errors; under the same SQL version and parameters, matched manual SQL exactly on **18 rows across all 19 fields**.
- Automated recurring A/B-test analysis from requirement parsing through multi-region execution and result delivery, reducing turnaround from **30–45 min to 2–3 min**; built validation utilities and an AI-readable knowledge base for historical analyses, business logic, metrics, methods, and best practices.

Jinken Vocational and Technical College

Mar. 2023 – Oct. 2025

Instructor & AI Technical Lead

China

- Designed and delivered Python, applied ML, and generative-AI curricula for **300+ students**; created hands-on labs, assessments, and project rubrics.
- Led STEM robotics initiatives and mentored an autonomous indoor-UAV team using ROS, scenario decomposition, and obstacle avoidance to win **Third Prize at the RAICOM 2024 National Finals**.

Selected Projects

Multi-Agent Literature Monitoring System | *Python, Google ADK, ChromaDB, SQLite, Langfuse*

Apr. – May 2026

- Architected the orchestrator for a team-built research-monitoring agent using **SequentialAgent + LoopAgent**, a critic-revision loop, and two human-in-the-loop checkpoints; designed typed schemas, dual storage, tracing, prompts, and guardrails.
- Delivered approximately **3,990 LOC in owned modules** across orchestration, storage, observability, and guardrails; passed **18/18 smoke tests** and the full pytest suite.

Hand-Coded Adversarial Agent | *Pure Python, standard library only*

May 2026

- Converted partial game observations into a transparent hierarchical policy for movement, aiming, and combat; iterated through black-box behavioral analysis, seeded tournaments, ablations, and systematic failure logging.
- **Tied for 1st of 6 agents** in battle royale and won **43% of 1v1 matches** against a neural-network opponent trained for 70M steps, without model training or external dependencies.

Sim2Real Calorie Prediction | *Python, scikit-learn, physics-informed ML*

2026

- Built a Sim2Real pipeline combining MET-based physical relabeling, explicit feature engineering, and data expansion from **973 to 20K samples**; evaluated generalization on **15K unseen real-world records**.
- Improved external R^2 from **-14.38** for a baseline random forest to **0.79** with linear regression, showing that distribution alignment and physics-grounded features outperformed added model complexity.

Honors

Huawei ICT Competition Global Final | Cloud Track, Global Third Prize (Top 5%)

Jun. 2022

Technical Skills

Languages: Python, SQL (Presto/Trino, Hive, MySQL), C++, Shell

Agent Systems: Google ADK, context engineering, Agent Skills/runbooks, Text-to-SQL, RAG, workflow automation

Reliability & Data: LLM evaluation, failure-mode analysis, guardrails, PyTorch, Transformers, PySpark, SQLite, Langfuse

Engineering: Git, Linux, ROS **Certifications:** Software Designer (PRC), Higher Education Teacher's Certificate in AI (PRC), IELTS 6.5